Prolog program for implementing the Towers of Hanoi

Aim

To write and execute a **Prolog program to implement the Towers of Hanoi** problem using recursion.

Algorithm

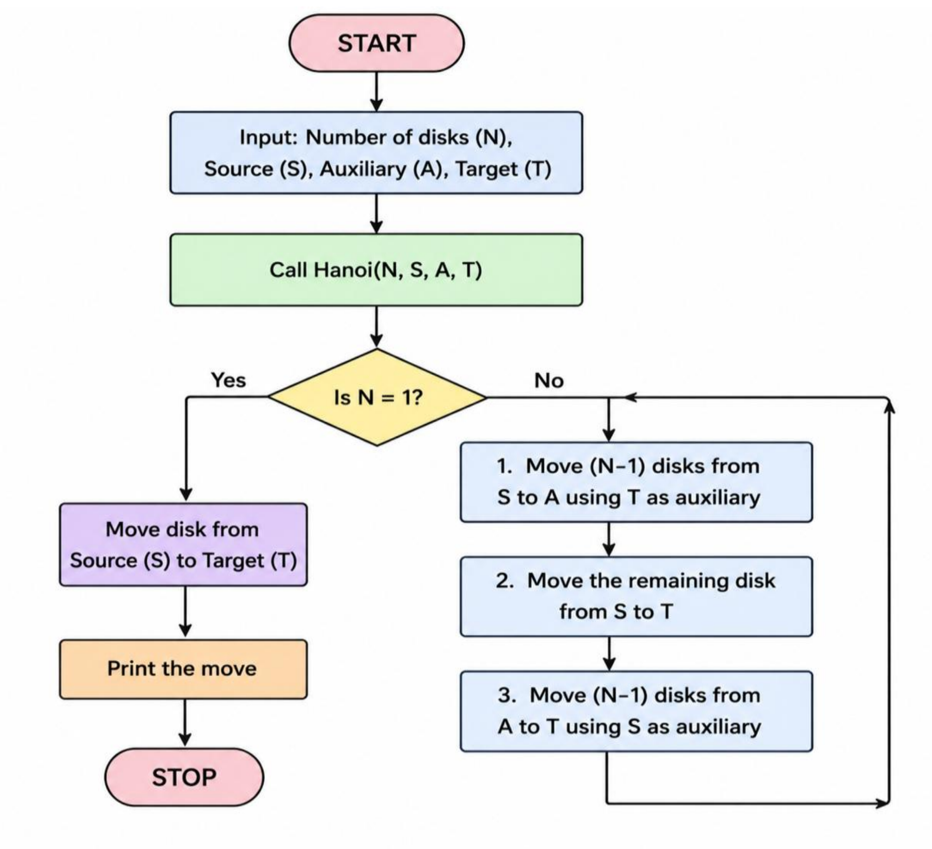
1. Start the program.
2. If N = 1, move the disk directly from the source peg to the target peg.
3. If N > 1, move N-1 disks from the source peg to the auxiliary peg.
4. Move the remaining largest disk from the source peg to the target peg.
5. Move the N-1 disks from the auxiliary peg to the target peg.
6. Print each disk movement using write/1 and nl/0.
7. Repeat the steps recursively until all disks are moved.
8. Stop.

Program

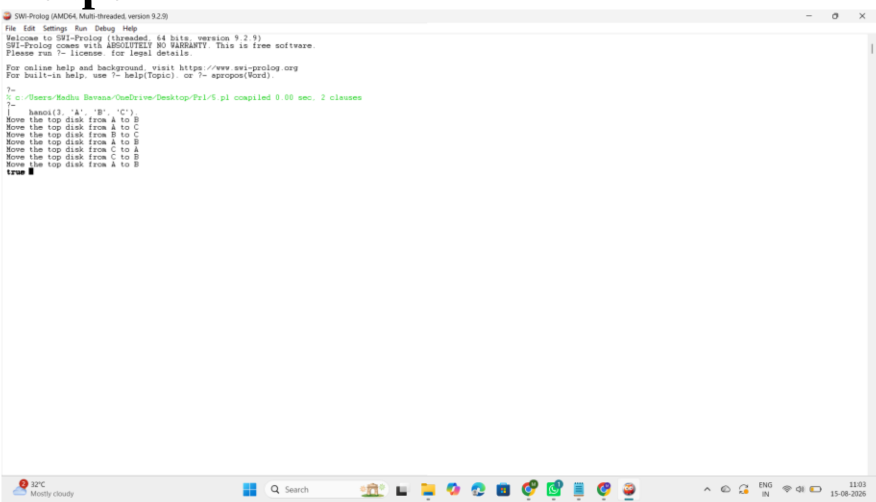
```
% Program to implement Towers of Hanoi

hanoi(1, Source, Target, _) :-
    write('Move the top disk from '),
    write(Source),
    write(' to '),
    write(Target),
    nl.

hanoi(N, Source, Target, Auxiliary) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, Source, Auxiliary, Target),
    write('Move the top disk from '),
```

```
write(Source),
write(' to '),
write(Target),
nl,
hanoi(N1, Auxiliary, Target, Source).
```

Flow chart



Output



Result

Thus, the **Prolog program for implementing the Towers of Hanoi was successfully executed**, and the required sequence of moves was obtained correctly.